Fundamentals of Derivatives Trading Strategies

Group 1: Rolling Futures Strategy

May 2026

Executive Summary

Project Objective

- Optimize the rolling window for E-Mini S&P 500 futures.
- Hedging a portfolio of the ten largest S&P 500 companies.
- Compare a **Baseline Strategy** (mechanical rolling) vs. an **Optimized Strategy** (signal-driven dynamic rolling).

Strategy Design & Market Exposure

Portfolio Construction

- Top 10 S&P 500 companies by market cap (Apple, Microsoft, etc.).
- Equal weight approach (10% each).

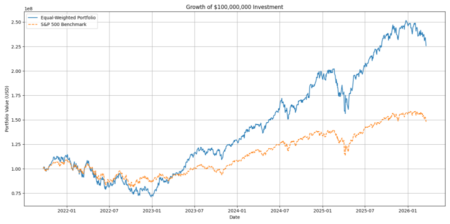

Figure 1: Cumulative Portfolio Return

Portfolio Performance Summary

Metric	EqualWeight	S&P500
Final Portfolio Value	$225.76M	$148.02M
Annualized Return	0.21	0.10
Annualized Volatility	0.24	0.17
Sharpe Ratio	0.87	0.60
Max Return	0.12	0.10
Min Return	-0.06	-0.06
Max Drawdown	-0.38	-0.38
Skewness	0.26	0.17
Kurtosis	4.76	7.03

Table 1: Equal Weight Portfolio vs. S&P 500

Strategy Design & Market Exposure

Market Exposure & Hedging

- 60-day rolling beta calculated:

$$\beta_{rolling} = \frac{Cov(R_p, R_m)}{Var(R_m)}$$

- Average beta ≈ 1.288.
- Target different levels of β by selling S&P 500 futures.

Strategy Design & Market Exposure

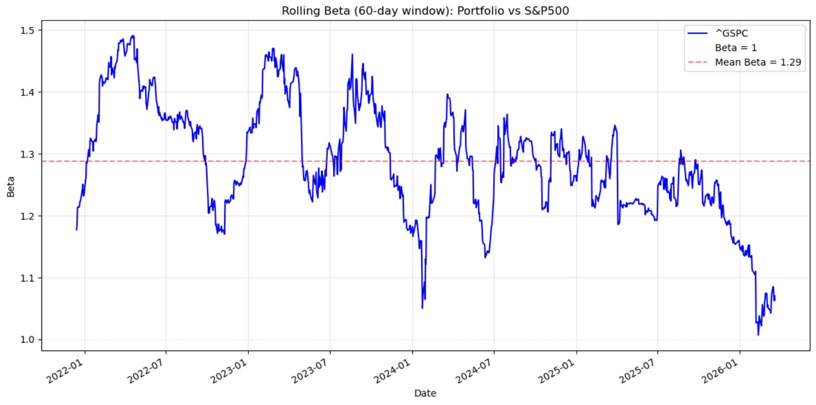

Figure 2: Rolling *Beta* over time

The E-Mini S&P 500 Futures Contract

Feature	Detail
Multiplier	\$50 per index point
Notional value	$\sim$\$250,000 – \$300,000
Minimum tick	0.25 index points = \$12.50
Trading hours	Nearly 24h on CME Globex
Expiry cycle	Quarterly (H, M, U, Z)
Settlement	Cash-settled, 9:30 a.m. ET on the 3rd Friday
Transaction tax	None

Why it matters

Each contract has a **finite life**. Maintaining continuous exposure requires periodically closing the expiring contract and opening the next one: **the roll**.

Theoretical Framework

The Cost-of-Carry Model

$$F_0 = S_0 \cdot e^{(r-q)\cdot T}$$

S_0	r	q
Spot index level (Yahoo Finance)	Risk-free rate (CME Term SOFR)	Dividend yield (Bloomberg IN060)

Contango ($r > q$)
Far contract > Near contract

Rolling earns money for short holder

Dominant from July 2022

Backwardation ($q > r$)
Far contract < Near contract

Rolling costs money for short holder

Late 2021 – mid 2022

Theoretical Framework

Time-to-Expiry

$$T_{c,d} = \frac{\text{Expiry}_c - d}{365}$$

- T is the number of calendar days remaining until expiry, divided by 365
- Defined only during each contract's active trading window

Why T matters
T appears **twice** in the fair value formula: directly, and indirectly through the interpolated risk-free rate.

Interpolating the Risk-Free Rate

Why CME Term SOFR?

- Forward-looking, collateralised, matched to the futures horizon
- Standard benchmark for margined derivatives — not overnight SOFR

The interpolation problem: T almost never coincides with a standard quoted tenor

Why a Natural Cubic Spline?

- Fits a **third-order polynomial** to each segment between adjacent tenor nodes
- **Exact fit** at every observed rate — actual borrowing costs matched precisely

Estimating the Dividend Yield

- S_0 and r are directly observable — q must be **estimated**

Source: Bloomberg `IN060` **(**`IDX_EST_DVD_CURR_YR`**)**

- Index-level estimated annual dividends, aggregated from constituent estimates
- Bloomberg does not provide constituent weights or dividend swap data
- Point-in-time history: no look-ahead bias in the backtest
- Updated continuously as new information arrives

$$q_d = \ln(1 + \text{Est. Dividend Yield}_d)$$

Mean $\approx 1.71\%$ Min $\approx 1.14\%$ Max $\approx 2.79\%$

Theoretical Framework

Fair Value and Basis: Measuring Mispricing

Fair Value — the synthesis of the full pipeline:

$$\text{Fair}_{c,d} = S_{0,d} \cdot \exp((r_{c,d} - q_d) \cdot T_{c,d})$$

Computed in vectorised form across all 18 contracts simultaneously.

Basis — daily diagnostic of mispricing:

$$\text{Basis}_{c,d} = \text{Market Price}_{c,d} - \text{Fair Value}_{c,d}$$

Positive basis
Contract trades **rich**

Good moment to re-short the deferred contract

Negative basis
Contract trades **cheap**

Less favourable timing to roll

From Basis to Signal: The Rolling Z-Score

$$z_d = \frac{\text{Basis}_d - \mu_{d,N}}{\sigma_{d,N}}$$

- Normalises the basis over a rolling N-day window
- Comparable across regimes with different absolute spread levels
- $z = +2/-2$: basis is 2 standard deviations above/below recent average — contract anomalously rich/cheap

Signal Roll
$z >$ threshold

$\Rightarrow$ Roll early, capture mispricing

Forced Roll
No signal fired

$\Rightarrow$ Roll anyway — never hold to settlement

Baseline Strategy: Systematic Rolling

- **Objective:** Benchmark for evaluating more sophisticated strategies
- **Approach:** Purely mechanical, ignores all market information
- **Position:** Constant short – one contract, always in the front month
- **Realistic costs:**
  - $50 contract multiplier (CME Group)
  - $2.50 per trade in transaction costs

Baseline Strategy: Systematic Rolling

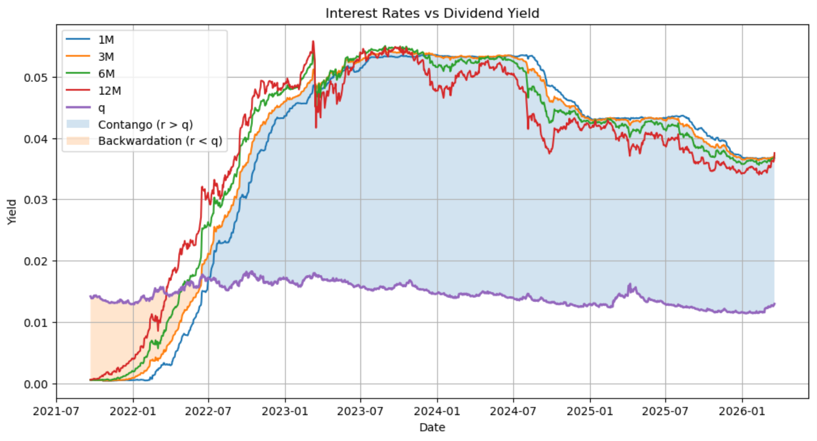

Figure 3: Interest Rate "r" versus Dividend Yield "q"

Baseline Strategy: Systematic Rolling

Roll Day	Total Loss	Market Spread	Fair Spread	Diff.
T-10	−78,166.00	689.75	558.62	131.12
T-9	−78,354.00	686.00	553.08	132.91
T-8	−78,429.00	684.50	552.24	132.25
T-7	−78,341.00	686.25	556.14	130.10
T-6	−78,329.00	686.50	557.97	128.52
T-5	−77,804.00	697.00	556.13	140.86
T-4	−77,466.00	703.75	557.60	146.14
T-3	−77,579.00	701.50	555.45	146.04
T-2	−78,241.00	688.25	557.92	130.32
T-1	−80,702.00	639.04	557.79	81.24

Table 2: Static Rolling Analysis (T = days before expiry)

Baseline Strategy: Systematic Rolling

- **Three weaknesses:**
  - **Static decision:** Ignores market conditions, pricing variations, fair value signals
  - **Marginal improvements:** T-10 to T-1 produces tiny differences; calendar adjustment is not enough
  - **Inefficiency ignored:** The +80 to +145 point gap is completely missed; rolls mechanically regardless of roll yield
- Optimizing a fixed schedule is not enough $\rightarrow$ we need a **dynamic approach**

Baseline vs. Optimized Strategy

Baseline Strategy (Systematic)

- Mechanical roll based strictly on time-to-maturity.
- Delivers slightly negative performance (-0.08% total return).
- Fails to exploit market inefficiencies.

Optimized Strategy (Dynamic)

- Opportunistic rolling based on mispricing (Basis Z-Score).
- Triggers early roll if z-score exceeds threshold, capturing rich spreads.
- Defaults to a forced roll if no signal triggers before expiry **(at $T - 3$)**.

Optimized Strategy: Dynamic Rolling

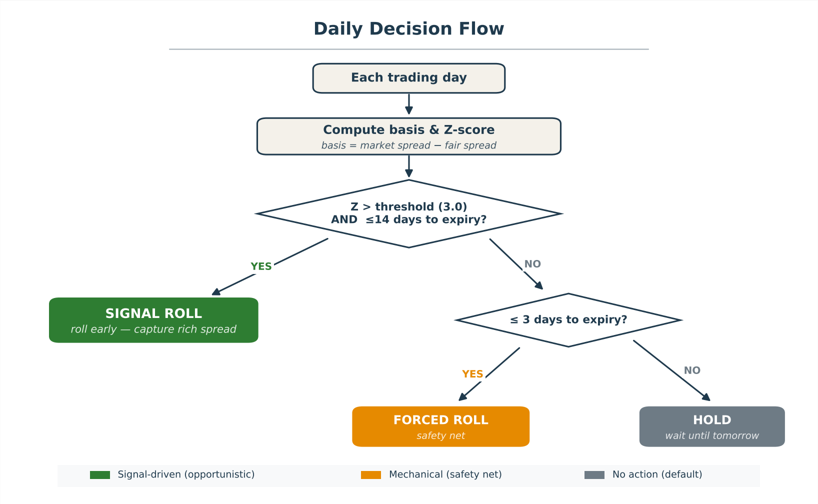

Figure 4: Two-tier roll decision logic: signal-driven roll when Z-score exceeds threshold, mechanical roll as safety net

Optimized Strategy: Dynamic Rolling

Parameter Selection

- Evaluated combinations of Look-back ($N \in \{10, 20, 30\}$ days) and Threshold ($z \in \{2.0, 2.5, 3.0\}$).
- **Optimal Region:** Short look-back + High threshold.
- Adapts quickly to regime changes while filtering out noise.

Results

- 8 out of 9 optimized configurations beat the best baseline.
- Best configuration improved spread capture by 4.22%.

Optimized Strategy: Dynamic Rolling

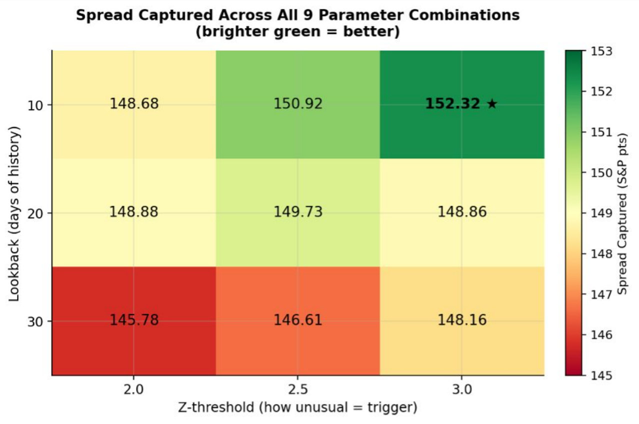

Figure 5: Spread captured for each combination of look-back window and Z-threshold

Optimized Strategy: Dynamic Rolling

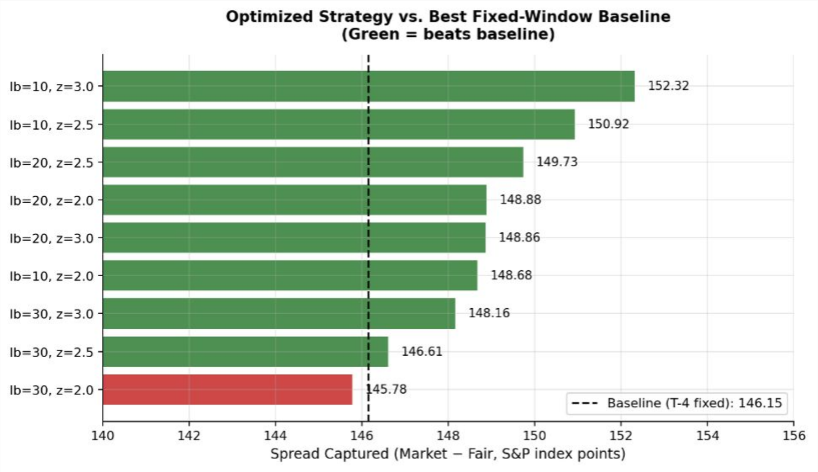

Figure 6: Spread captured by every parameter combination of the optimized strategy, compared against the best fixed-window baseline

Strategy Design & Market Exposure

Implementation

- Parameters: 20-day look-back, 2.0 z-score threshold.
- Dynamic hedge ratio h used to adjust number of contracts:

$$N_t = h \cdot \frac{\beta_{t-1} \cdot I_{t-1}}{F_t \cdot M}$$

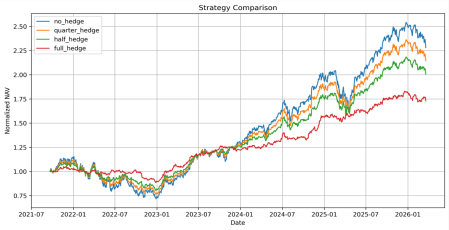

Figure 7: Cumulative returns for different level of hedged portfolio

Hedging Strategy Performance Metrics

Strategy	Ann. Return	Ann. Vol.	Sharpe	Calmar	Max DD	Win Rate
no_hedge	0.202	0.242	0.880	0.532	-0.380	0.539
quarter_hedge	0.185	0.191	0.985	0.571	-0.324	0.532
half_hedge	0.168	0.145	1.141	0.628	-0.267	0.523
full_hedge	0.130	0.096	1.318	0.807	-0.161	0.539

Table 3: Performance by Hedge Ratio

Risk Distribution & Cost Analysis

Strategy	Max Ret.	Min Ret.	Skew.	Kurt.	Final NAV	Trans. Cost
no_hedge	0.120	-0.061	0.254	4.757	$227.5M	$200k
quarter_hedge	0.090	-0.046	0.207	3.723	$213.8M	$215k
half_hedge	0.060	-0.039	0.159	2.520	$200.0M	$230k
full_hedge	0.033	-0.025	0.040	1.740	$172.5M	$260k

Table 4: Risk Metrics & Transaction Costs

Back-Test Results

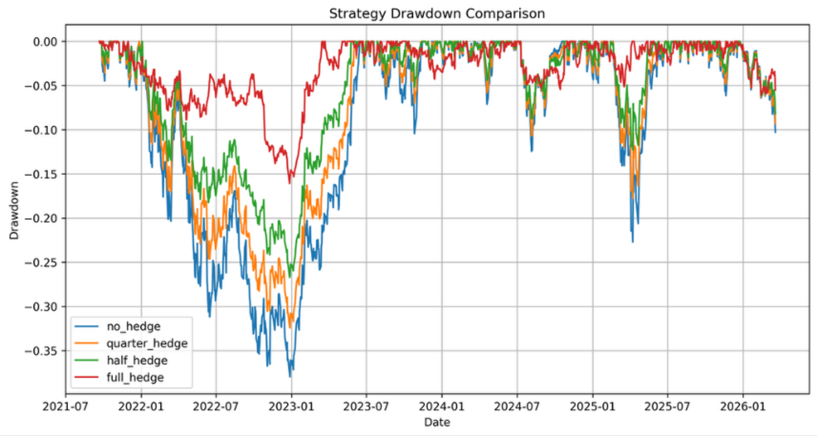

Figure 8: Max Drawdown by Hedging Strategy

Back-Test Results

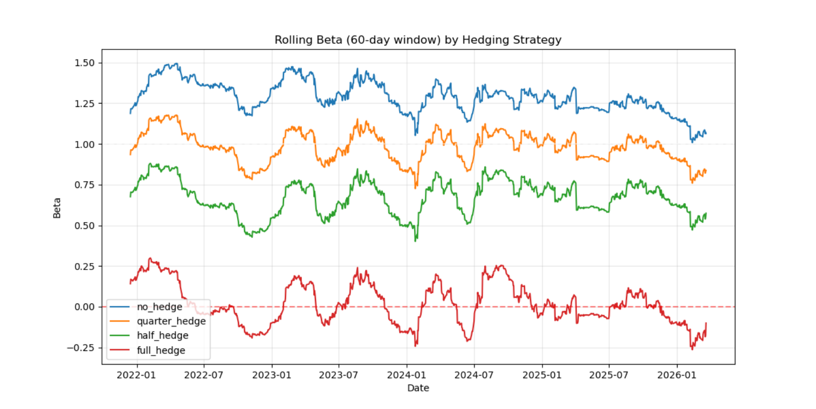

Figure 9: Rolling Beta by Hedging Strategy

Key Takeaways

- **Best Sharpe Ratio:** 1.318 (full hedge)
- **Best Calmar Ratio:** 0.807 (full hedge)
- **Highest Return:** 20.2% (no hedge)
- **Optimized rolling improved spread capture by +4.22%**
- **8 out of 9 optimized configurations beat the fixed baseline**

Conclusion

- Mechanical rolling leaves potential alpha on the table.
- Utilizing the cost-of-carry framework to identify anomalous basis deviations allows for superior spread capture.
- Dynamic hedging provides structural protection during market shocks, resulting in superior risk-adjusted performance.

References

Athari, M., Naka, A., and Noman, A. (2021). "Predicting equity premium with adjusted dividend-price ratio: the USA and international evidence." *Review of Accounting and Finance*, 20(3/4), pp. 217–247.

Brown, L., Call, A., Clement, M., and Sharp, N. (2002). "Inside the "black box" of sell-side financial analysts." *Journal of Accounting Research*, 53(1), pp. 1–47.